

Homestay Phú Nhuận

Automated Booking Website • Portfolio Case Study

An automated homestay booking platform that lets customers check real-time availability, book online, pay by QR and receive door access codes with minimal owner interaction.



Core differentiator:

Guests can book at night, pay online and receive check-in information immediately — so the owner can sleep without monitoring messages or the hotline.



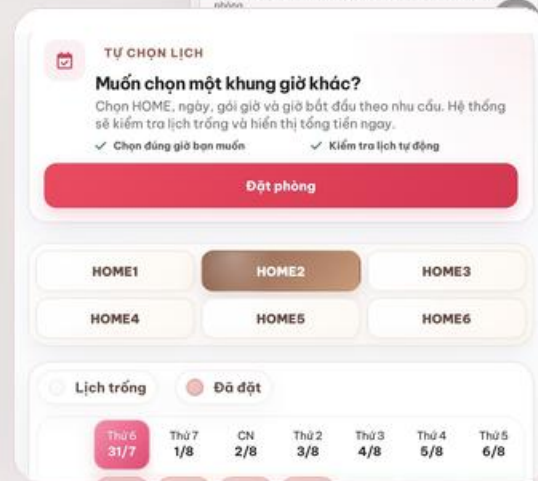
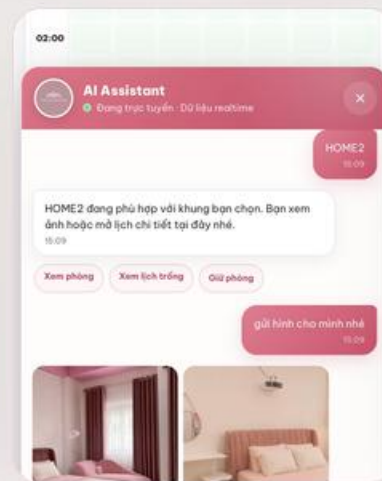
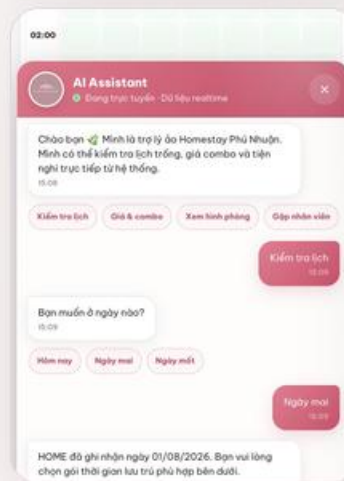
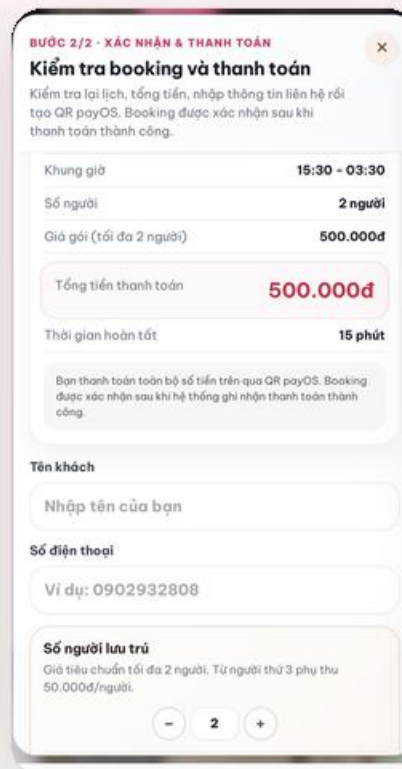
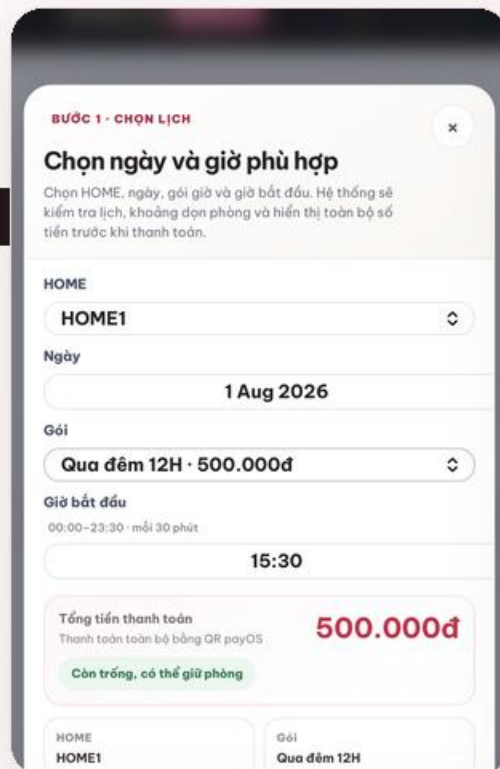
Realtime
availability



Automated
online booking



TTLock
self check-in



Business problem solved by the website

From manual operations to a 24/7 self-service booking workflow.

Common manual model

- 1 Guests must call or message to ask for availability.
- 2 The owner checks Excel or manually verifies the calendar.
- 3 Late-night guests may be missed if the owner is asleep.
- 4 Payment confirmation and door codes are handled manually.
- 5 Higher risk of wrong slots, double-booking or missing information.



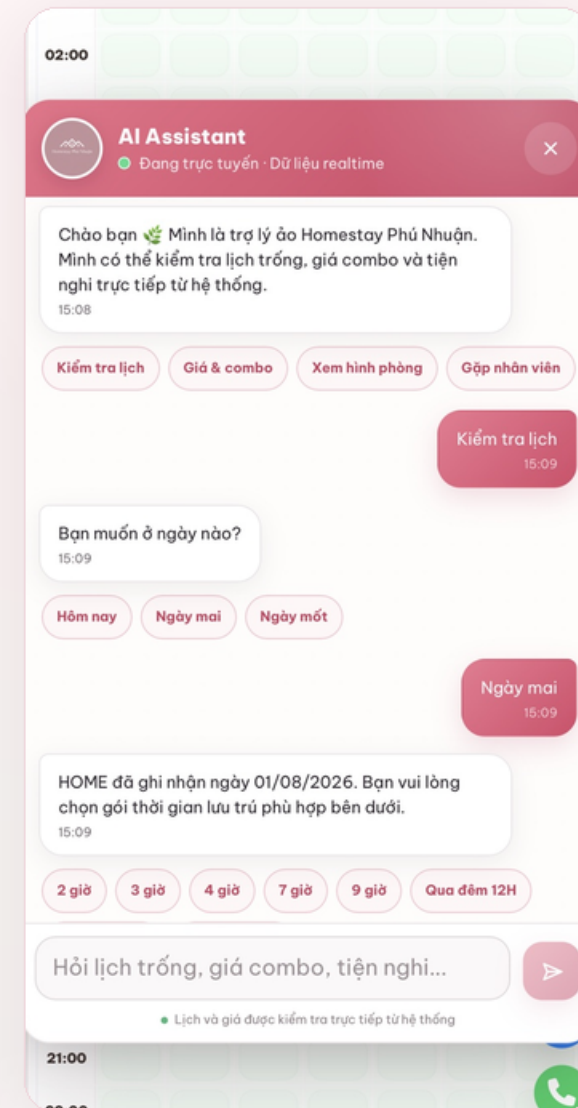
Homestay Phú Nhuận solution

- 1 Guests check live room availability by themselves.
- 2 The system validates whether the requested time slot can be booked.
- 3 Guests book, pay and receive check-in information automatically.
- 4 Night bookings can be completed without the owner being online.
- 5 Bookings and "Talk to staff" requests are pushed to a Telegram group.



Biggest business value: the owner can sleep at night and does not need to monitor the hotline, while the system still captures overnight booking demand.

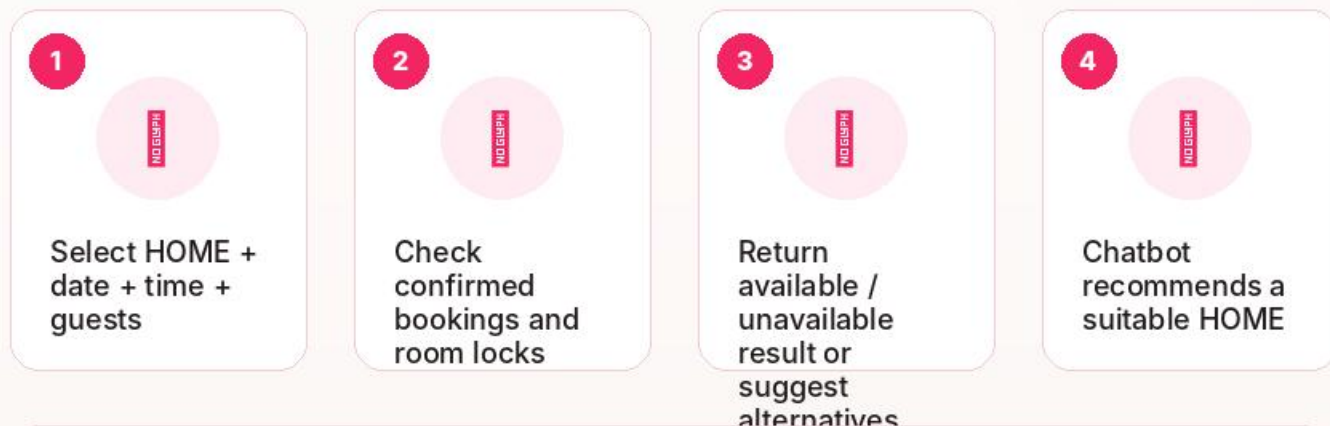
Problem → Solution transformation



Homestay Phú Nhuận • Portfolio Case Study

Advantage 1— Realtime availability + automatic slot validation

Guests do not need to ask manually. The system checks whether the selected HOME, date, time and guest count can be booked.



Unlike the common approach



Many homestays still send an Excel sheet or ask customers to message for open slots.



Homestay Phú Nhuận lets customers self-check, while the chatbot supports them automatically.



Live sync

Every update is reflected immediately.



Higher accuracy

Bookings and room locks are checked together.



Smooth experience

Find a slot and book in a few clicks.

TỰ CHỌN LỊCH

Muốn chọn một khung giờ khác?
Chọn HOME, ngày, gói giờ và giờ bắt đầu theo nhu cầu. Hệ thống sẽ kiểm tra lịch trống và hiển thị tổng tiền ngay.

✓ Chọn đúng giờ bạn muốn ✓ Kiểm tra lịch tự động

Đặt phòng

HOME1 HOME2 HOME3
HOME4 HOME5 HOME6

Lịch trống Đã đặt

	Thứ 6 31/7	Thứ 7 1/8	CN 2/8	Thứ 2 3/8	Thứ 3 4/8	Thứ 4 5/8	Thứ 5 6/8
00:00							
01:00							
02:00							
03:00							
04:00							
05:00							
06:00							
07:00							

Zalo

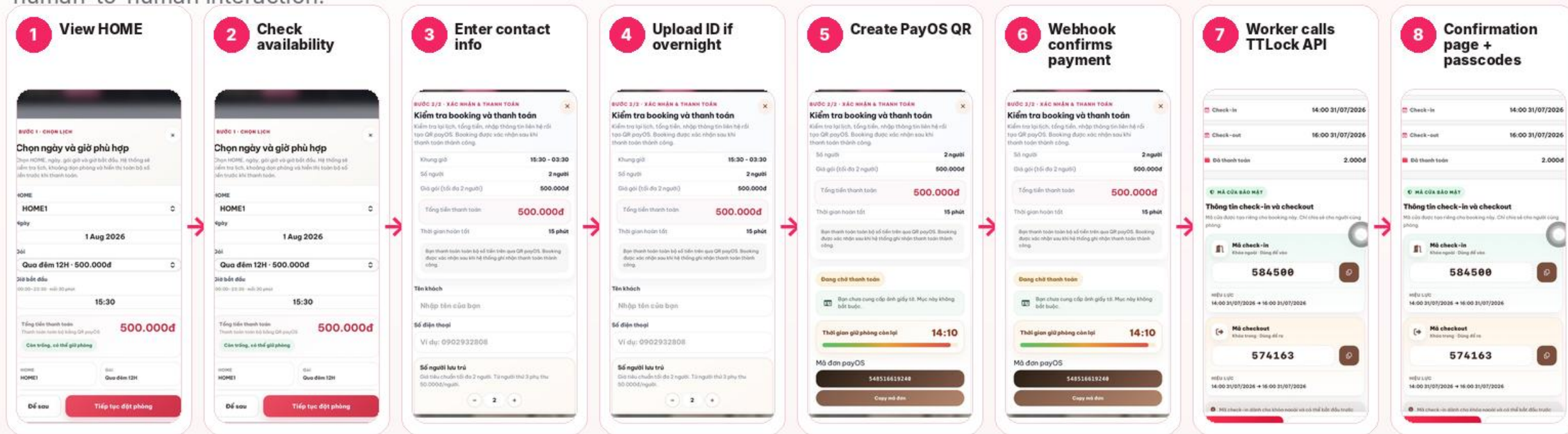
Realtime availability

Slot validation

Chatbot support

Advantage 2 — Fully automated online booking flow

From viewing a HOME to receiving access codes, the flow requires almost no human-to-human interaction.



After successful payment, the guest immediately receives booking details, directions to the HOME and time-bound check-in / check-out passcodes. The confirmation page remains accessible until checkout.



Result: booking, payment and entry can be completed even when the owner is not online.

Night booking scenario: pay online and check in immediately

Designed for urgent, after-hours demand — while keeping owner operations under control.



AUTOMATIC OWNER SUPPORT VIA TELEGRAM



Operational meaning

The website acts like a 24/7 digital front desk. The flow does not depend on whether the owner is awake and eliminates back-and-forth confirmation messages.

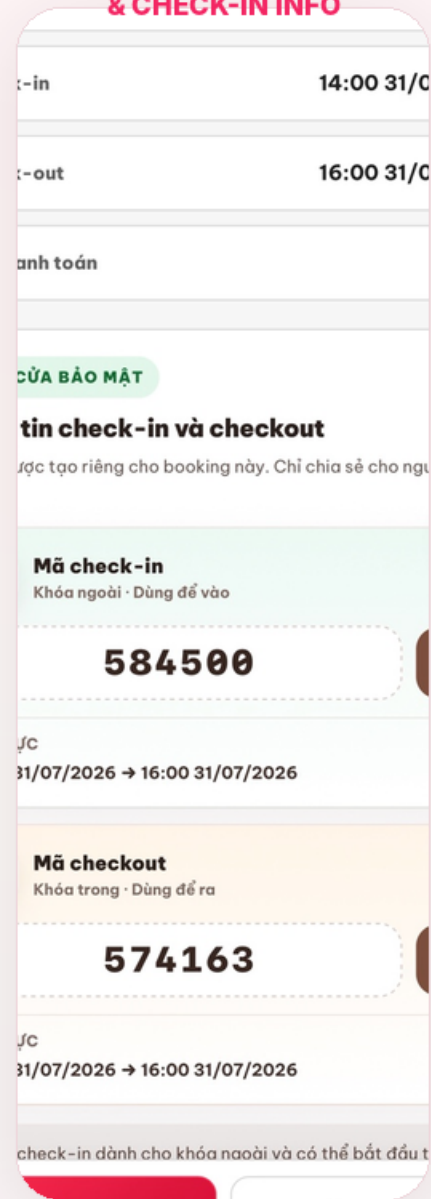
Smart lock time control

Check-in codes become valid from the scheduled time to prevent early access. Checkout codes are limited to the checkout window to avoid overstay.

WAITING FOR PAYMENT

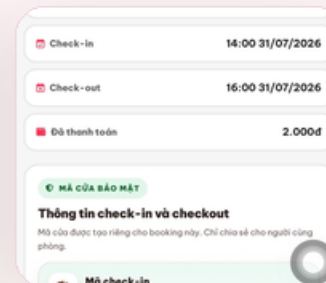
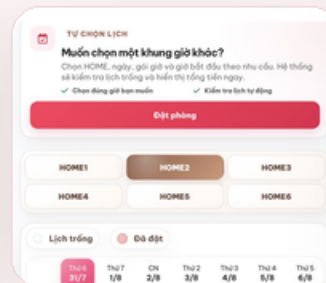
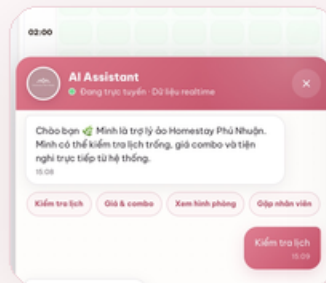


BOOKING CONFIRMED & CHECK-IN INFO



TECH & system architecture

A serverless booking workflow built with a lightweight frontend, centralized realtime data and external service integrations.



1. Frontend / Customer UI

- HTML5, CSS3, JavaScript ES Modules
- Responsive customer website
- Booking modal & confirmation page
- Chatbot widget



2. Serverless backend

- Node.js on Vercel Serverless Functions
- Availability validation
- Pricing / combo rules
- PayOS webhook handling



3. Realtime data layer

- Firebase Realtime Database
- bookingsByMonth & room locks
- roomPricing / combos
- homes & businessProfile
- payments & customers



4. Integrations

- PayOS QR payments
- TTLock smart lock API
- Meta Messenger + Gemini
- Telegram Bot alerts
- Cloudinary signed uploads



5. Engineering controls

- Server-side secrets
- Webhook verification
- Booking + lock consistency
- Auth & database rules
- Graceful failure handling

Customer

Visits website, checks availability and books online.



System

Validates slots, calculates price, processes payment and generates access codes.



Owner / Admin

Receives Telegram alerts, manages exceptions and supports guests.

Outcome & advantages over common homestay operations

The value is not just a nice website — it is end-to-end automation of the booking workflow.

01

1

End-to-end workflow automation

Availability check, payment and passcode delivery all run through a digital workflow.

02

24

24/7 operation

Guests can book at night or when the owner is not online.

03

1

Less manual work for the owner

Fewer repetitive messages about availability, price, payment and door codes.

04

✓

More private guest experience

Guests choose time, pay online and self check-in / check-out quickly.

★

Key portfolio takeaway: the owner can sleep, while the system continues selling rooms, sending alerts and enforcing the correct booking rules.

Compared with typical homestays in Ho Chi Minh City

	Common model	Homestay Phú Nhuận
View availability	Manual / inbox	Realtime, automated
Validate requested slot	Depends on owner reply	Instant slot validation
Hold room during payment	Manual transfer confirmation	15-minute room lock
Confirm payment	Transfer screenshot	PayOS webhook
Send door code	Manual code sharing	TTLock passcodes
Night booking	Wait for owner response	24/7 self-service
Support request	Scattered chats / hotline	Telegram group tracking

Telegram Bot & support automation

No hotline monitoring, but no missed bookings or support requests.

Website chatbot

The AI Assistant includes a "Talk to staff" path and can check availability / pricing directly from the system.

1 Booking via website

Guest pays or holds a room online; the system sends a Telegram alert.

2 Guest taps "Talk to staff"

The request is forwarded to the Telegram group for owner/admin follow-up.

3 Owner does not need to stay online overnight

Booking control and customer support remain visible in one group.

Telegram operations group

New website booking

HOME2 • Overnight 12H • 500,000 VND Payment confirmed via PayOS

Door passcodes generated

Check-in: 584500 • Checkout: 574163 Valid according to the booking time window

Guest wants to talk to staff

Request forwarded to the group so admin can track and respond.

Realtime data principle

The website and chatbot do not respond with static content. Every check reads the latest schedule, pricing, combo packages and HOME information from Firebase in real time.